Hypothesis Testing and the Null Hypothesis.

Hypothesis Testing

Part of Research → Accept / Reject

Based on Data

→ Statistical method used to evaluate a claim / hypothesis about population parameter

- Comparing sample data to hypothesis about a population parameter to determine whether there is enough evidence to reject or accept the hypothesis.
- Without hypothesis test, you risk drawing wrong conclusion & making bad decision.

(*) Null hypothesis → Proof ?
Represent (H0)

→ baseline, default assumption
→ there is no significant difference or effect.

(2) Alternate hypothesis – (H_1)

→ Logically opposite of null hypothesis – there is significant difference or effect.

These two claims need to be mutually exclusive.

eg – claim : Avg. height of India is more than 170cm.

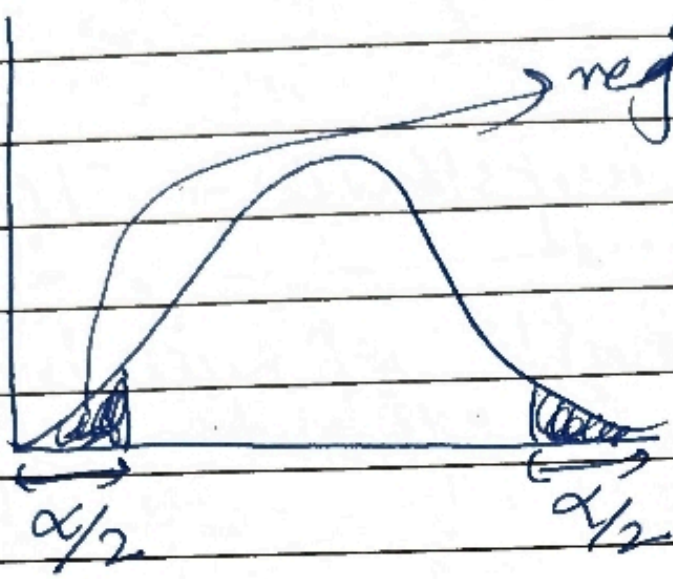
Null hypothesis – H_0 : $\mu = 170cm$ / $\mu_0 \leq 170$

Alternate hypothesis – H_1 : $\mu > 170cm$.

(*) Score / level

Level of Significance (α) :-

Predetermined threshold – rejection region of distribution



P- value - strength of evidence against H_0

P value compare - α

Decision Rule :-

If P. value is less than α. reject the H_0.

$P < \alpha$, H_0 reject.